

Motivation and Career Development Statement

Marie Skłodowska-Curie Postdoctoral Fellowship 2026 — DAIRYTWIN

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Proposed host: Technische Universität Dresden — ScaDS.AI Dresden/Leipzig

An engineer who chose agriculture

I was trained as a computer engineer at the Czech Technical University, and I spent the first decade of my career writing software for banks, payment networks, and public-sector information systems — EESSI for European social-security exchange, citizen-facing tools at the Ministry, security and privacy services at Avast. The work was useful, but at some point I noticed that the people who most needed good software — farmers, veterinarians, family-run businesses outside Prague — were the people the technology industry was building for last. That observation moved me out of commercial software and into a Ph.D. in Systems Engineering at the Czech University of Life Sciences, which I defended in 2025. Research is the part of this trajectory I am most enthusiastic about: I now hold a postdoctoral track in which I get to study, full time, how artificial intelligence and operations research can become a quiet, dependable assistant for a dairy farmer at five in the morning — and I would rather do that than anything else.

The MSCA Postdoctoral Fellowship is the right next step in this trajectory. It is the European programme that explicitly asks two questions at once — *is your science excellent enough?* and *is the European Research Area genuinely stronger because of your work?* I have spent the last four years building my answer to both.

What I do, and why dairy

My doctoral thesis built data-driven decision-support systems and predictive models for dairy herd health: Markov chains for disease transitions, a herd-health DSS published in the *Czech Journal of Animal Science*, and disease-scoring tools that farmers can use without a data scientist in the room. The work earned the Dean's First Prize for the best scientific publication at FEM CULS in 2024 and, more importantly, has been picked up by people who actually run cows. In the same year, I was a finalist of the Falling Walls Lab Czech Republic with a talk titled *Breaking the Wall of the Dairy Farmer AI Assistant* — a deliberate attempt to translate the work into language a non-specialist audience could follow. I now hold parallel appointments as Assistant Professor at CULS and Junior Researcher at the Institute of Information Theory and Automation of the Czech Academy of Sciences, which gives me one foot in agricultural reality and the other in mathematical rigour.

Dairy matters in the European Union for reasons that should be central to the next decade of EU agricultural policy. The EU produces roughly 155 million tonnes of cow milk per year from approximately 20 million dairy cows on more than 1.4 million farms; subclinical mastitis alone is estimated to cost the EU dairy sector on the order of €1.5 billion annually, and dairy carries a disproportionate share of EU agricultural methane. At the same time, dairy is one of the few remaining sectors where small and medium family farms still compete with large operations, and the gap between data-rich and data-poor producers is widening fast. A decision-support tool that

genuinely respects the farmer's expertise is therefore not only a technical artefact — it is a small instrument of fairness, of welfare, and of the Farm-to-Fork strategy at the producer level.

Open science: the line I will not cross

The core commitment of this fellowship — and the line that aligns me most directly with MSCA values — is that every line of code, every dataset transformation and every model produced under DAIRYTWIN will be *open by default, auditable by a non-specialist and welfare-first by design*. There is real temptation in agricultural AI to optimise yield at the expense of the animal, or to lock decision logic behind proprietary platforms that small farms cannot afford. I have turned down freelance offers for both. The line is not difficult to hold; it just has to be held publicly. The Open Science Plan that I will maintain throughout the fellowship — gold open-access publications, MIT-licensed code, FAIR data, preprints from M1 — is not bureaucratic compliance; it is the project's organising principle.

This is also why I am applying to a postdoctoral fellowship rather than to industry. The right venue for a tool that small EU farms and extension veterinarians can use is a public-good research environment, working alongside Europe's strongest AI methodology groups. The MSCA programme aligns those incentives in a way that no commercial post would.

Why this host, why the European Research Area, why now

I have studied the dairy decision-support and digital-twin literature across the EU, U.S. and Australasian communities carefully. The methodological depth required to turn a cow-level prototype into a scalable, trustworthy, explainable system sits in a small number of European groups, and **ScaDS.AI Dresden/Leipzig at TU Dresden** is in my view the strongest single fit. ScaDS.AI is one of six German national competence centres for AI, with a remit that covers exactly the three pillars on which DAIRYTWIN is built: scalable methodology, AI for natural sciences, and trustworthy and explainable AI. Hosting the project there means the cow-level state estimator, the constrained herd-prioritisation policy and the farmer-facing explanation layer are built next to people whose daily work is making large-scale AI methods reliable and accountable.

Twelve months is the right size for this commitment. It is long enough to deliver a working hybrid digital-twin prototype, a reproducible open benchmark, a co-authored peer-reviewed publication, and a structured secondment to a complementary EU dairy-science partner; it is short enough that my Ph.D. students in Prague do not lose continuity and that the return-phase integration into CULS and ÚTIA is realistic from day one.

What this fellowship will leave behind

I do not view an MSCA stay as something I receive — I view it as a shared output that should leave both sides of the European Research Area materially better off. The fellowship is built around two concrete artefacts that benefit the host institution and the Czech research base equally. The first is a working digital-twin prototype, hardened to a level at which a company or a start-up — in Germany, in the Czech Republic, or jointly — could adopt it as the basis of a deployment, in the spirit of a Horizon Europe project whose output is a usable technology rather than a closed report. The second is a peer-reviewed scientific publication on the design and validation of the

twin, with co-authorship for members of the host group secured from the outset of the stay. Both artefacts are vehicles for the broader purpose of the MSCA programme: to widen and deepen scientific cooperation across the European Research Area. The publication anchors a long-term research relationship between TU Dresden and CULS / ÚTIA; the prototype anchors a long-term applied-research relationship; together they give both institutions something concrete to build on after I return home — including a credible base for a follow-up Horizon Europe project or an ERC Starting Grant.

A short word about how I work

I am a builder by temperament. I prefer a working prototype over a clean slide. I have run a freelance web-development practice for over a decade alongside academic work — more than fifty delivered web systems — and I know how to ship under real-world constraints. I have an MBA, which means I can talk to funders, vendors and farm owners in their own language. Applied AI is also a personal enthusiasm of mine, not only a professional one — I build small AI tools in my free time, follow the field closely outside working hours, and keep returning to the question of how to make these models genuinely useful to people who are not engineers. I am at C1 in English and I am committed to reaching working competence in German during the fellowship. I am thirty-six, recent enough after my Ph.D. defence to qualify as a postdoctoral fellow but mature enough to know which projects are worth twelve months of my life. This one is.

Respectfully submitted,
Ing. Jan Saro, Ph.D., MBA